

Independent and Combined Effects of Dietary Weight Loss and Exercise on Leukocyte Telomere Length in Postmenopausal Women

Objective Investigate the effects of 12 months of dietary weight loss and/or aerobic exercise on leukocyte telomere length in postmenopausal women. **Design and Methods** 439 overweight or obese women (50–75 y) were randomized to: i) dietary weight loss (N=118); ii) aerobic exercise (N=117), iii) diet + exercise (N=117), or iv) control (N=87). The diet intervention was a group-based program with a 10% weight loss goal. The exercise intervention was 45 mins/day, 5 days/week of moderate-to-vigorous aerobic activity. Fasting blood samples were taken at baseline and 12 months. DNA was extracted from isolated leukocytes and telomere length was measured by quantitative-polymerase chain reaction (qPCR). Mean changes were compared between groups (intent-to-treat) using generalized estimating equations. **Results** Baseline telomere length was inversely associated with age ($r=-0.12$ $p<0.01$) and positively associated with maximal oxygen uptake ($r=0.11$, $p=0.03$), but not with BMI or %body fat. Change in telomere length was inversely correlated with baseline telomere length ($r=-0.47$, $p<0.0001$). No significant difference in leukocyte telomere length was detected in any intervention group compared to controls, nor was the magnitude of weight loss associated with telomere length at 12 months. **Conclusions** Twelve-months of dietary weight loss and exercise did not change telomere length in postmenopausal women.